Arista Access Node KOR-ACC-66 Routing Protocol Flap Resolution SOP

Node: KOR-ACC-66 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9227

Description

This Standard Operating Procedure (SOP) addresses the root cause of the Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain on Arista Access Node KOR-ACC-66 at the Koramangala Datacenter. The failure domain originates from the dynamic nature of routing protocols, which are designed to adapt to changing network conditions. BGP route churn can be triggered by various factors such as route updates, prefix deletions, or network topology changes. OSPF adjacency loss can occur due to link failures, neighbor relationship issues, or network partitioning. ISIS LSP flooding can be caused by topology changes, route updates, or hardware issues. In the context of Arista Access Node KOR-ACC-66, the root cause of the failure domain is likely to be related to network congestion, misconfiguration, or hardware malfunction. This SOP provides a step-by-step guide to diagnose and resolve the issue by leveraging Arista's proprietary CLI dialect and leveraging expert knowledge of routing protocol mechanics.

Pre-requisite Checks

- Verify the Koramangala Datacenter physical environment and ensure that all necessary personnel have access to the site.

- Check the status of redundant links and ensure they are functional.

- Verify that the Arista Access Node KOR-ACC-66 is powered on and its management interface is reachable.

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor relationships	High
OSPF adjacency loss due to link failures	Medium
ISIS LSP flooding caused by hardware issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 2, main routing table version is 2 BGP sending router ID 192.168.1.1 BGP state: established, up for 00:05:30 BGP table version 2, neighbor version 2 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd 10.1.1.1 65000 2 2 2 0 0 00:05:30 0 , Description: This command provides a summary view of BGP session information, including neighbor relationships, message reception and transmission, and routing table version. The output will help identify if any BGP neighbor relationships are misconfigured.
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```
Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface 192.168.1.2 1 FULL/ - 00:00:33 10.1.1.2 GigabitEthernet1/1 , Description: This command displays a list of OSPF neighbors, including their ID, priority, state, and dead timer. The output will help identify if any OSPF adjacency losses are occurring.

Command: show isis database, Expected Output: ISIS LSP Database: LSP ID Age Sequence Checksum 10.1.1.1.01 00:00:10 0x8000000 0x12345678 10.1.1.2.01 00:00:15 0x8000000 0x23456789 , Description: This command displays the ISIS LSP database, including LSP ID, age, sequence number, and checksum. The output will help identify if any ISIS LSP flooding is occurring.
```

Resolution Steps

Step 1: Step: Verify and correct BGP neighbor relationships, Command: show ip bgp summary | match Neighbor V, Description: Verify that all BGP neighbor relationships are correctly configured. If any misconfigurations are found, correct them and re-run the diagnostic command to verify the fix.

Step 2: Step: Verify and correct OSPF adjacency losses, Command: show ip ospf neighbor | match FULL/, Description: Verify that all OSPF adjacency losses have been resolved. If any losses are still occurring, identify and correct the root cause, such as link failures or network partitioning.

Step 3: Step: Verify and correct ISIS LSP flooding, Command: show isis database | match LSP ID, Description: Verify that all ISIS LSP flooding has been resolved. If any flooding is still occurring, identify and correct the root cause, such as hardware issues or topology changes.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 2, main routing table version is 2 BGP sending router ID 192.168.1.1 BGP state: established, up for 00:10:00 BGP table version 2, neighbor version 2 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd 10.1.1.1 65000 2 2 2 0 0 00:10:00 0 , Description: Verify that the BGP session is stable and all neighbor relationships are correct.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface 192.168.1.2 1 FULL/ - 00:00:33 10.1.1.2 GigabitEthernet1/1 , Description: Verify that all OSPF adjacency losses have been resolved.

Command: show isis database, Expected Output: ISIS LSP Database: LSP ID Age Sequence Checksum 10.1.1.1.01 00:00:10 0x8000000 0x12345678 10.1.1.2.01 00:00:15 0x8000000 0x23456789 , Description: Verify that all ISIS LSP flooding has been resolved.
```

Roll-back Procedures

- Step: Restore previous BGP configuration, Command: show ip bgp summary | match Neighbor V, Description: Restore the previous BGP configuration to resolve the issue.

- Step: Restore previous OSPF configuration, Command: show ip ospf neighbor | match FULL/, Description: Restore the previous OSPF configuration to resolve the issue.

- Step: Restore previous ISIS configuration, Command: show isis database | match LSP ID, Description: Restore the previous ISIS configuration to resolve the issue.